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Socioeconomic disparities in prehospital cardiac arrest outcomes: An analysis of the NEMSIS database

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ABSTRACT

Background: Socioeconomic disparities are engrained in the US healthcare system and may extend to the prehospital cardiac arrest setting where mortality is high.

Methods: Using the National Emergency Medical Services Information System (NEMSIS) database, 150,003 cases were analyzed comparing socioeconomic status and cardiac arrest outcomes. Cardiac arrest outcomes were measured by the percent of cases that achieved return of spontaneous circulation (ROSC) and the percent of cases in which ROSC occurred in the Emergency Department (ED) as opposed to a prehospital setting which was a proxy for the length of time spent in cardiac arrest. Chi-square tests checked for statistical significance and effect size was measured using Pearson's r values and linear regression coefficients.

Results: Comparing neighborhood poverty level and the percent of cardiac arrest cases that achieved ROSC resulted in a Pearson's r value of 0.9424 ($R^2 = 0.8881$, $p < 0.005$) and a linear regression coefficient of 2.088 ($p < 0.05$, $R^2 = 0.8881$, 95% CI [1.059, 3.117]) meaning for every interval increase in poverty, the chance of an individual in cardiac arrest achieving ROSC decreases 2.09%. Comparing neighborhood poverty level and the percent of ROSC cases that occurred in the ED yielded a Pearson's r value of 0.9005 ($R^2 = 0.8109$, $p < 0.05$) and a linear regression coefficient of 0.7701 ($p < 0.05$, $R^2 = 0.8109$, 95% CI [0.254, 1.286]) meaning for every interval increase in poverty, the chance that ROSC is delayed increases 0.77%.

Conclusions: Low income individuals in cardiac arrest have a statistically significant lower probability of achieving ROSC and a higher chance of delayed ROSC.

1. Introduction

Socioeconomic status (SES) remains a strong predictor of the health of an individual in the United States [1]. With life expectancy decreasing as SES declines, it is clear that health in the US is patterned along socioeconomic lines [2]. A key factor that contributes to the socioeconomic disparities found in health is the quality of care that the healthcare system provides. On average, low income individuals receive a worse quality of care from the US healthcare system than high income individuals [3]. One of the most important scenarios in which quality of care is critical is during cardiac arrest as it is the most common cause of death in the US leading to over 500,000 deaths per year [4,5]. Out of all cardiac arrests, over 400,000 occur in a prehospital setting with an over 94% fatality rate compared to a 76% fatality rate in a hospital setting [6,7]. This makes the prehospital setting a critical point for cardiac arrest care. The goal of this study is to determine if the socioeconomic disparities found in the overall US healthcare system continue to the prehospital cardiac arrest setting.

Prior studies on the differences in prehospital cardiac arrest outcomes between socioeconomic groups report inconsistent findings [8-10]. Furthermore, many of these studies lack a nationally inclusive sample which this study addresses through the use of the NEMSIS database [8,9]. In addition, it appears that prior studies that analyze socioeconomic disparities in prehospital cardiac arrest outcomes focus solely in measuring outcomes in the form of survival or ROSC rates. However, not all survivors of cardiac arrest are equal since significant differences are present in neurological function based on the length of time spent in cardiac arrest [11]. No prior study was found that analyzes socioeconomic disparities in the length of time spent in cardiac arrest until ROSC occurred. This study corrects for this by measuring cardiac arrest outcomes not only by survival rate but also by the neurological function of the survivors.

2. Materials and methods

This study used a retrospective cross-sectional analysis of EMS data from the 2017–2019 NEMSIS V3.5.0 database. NEMSIS is a voluntary

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Table 1
Incidence of return of spontaneous circulation (ROSC) based on geographic poverty rate.

Poverty percent	No ROSC	Yes ROSC	% No ROSC
≤5%	4668	4211	52.57%
6%–10%	16,130	12,253	56.83%
11%–15%	21,730	14,732	59.60%
16%–20%	17,317	11,849	59.38%
21%–25%	11,337	7456	60.33%
≥26%	18,446	9874	65.13%
Total	89,628	60,375	

Table 2
Location when return of spontaneous circulation (ROSC) occurred based on geographic poverty rate.

Poverty percent	At ED	Prior to ED	% at ED
≤5%	659	3522	15.65%
6%–10%	2203	10,050	17.98%
11%–15%	2730	12,002	18.53%
16%–20%	2292	9557	19.34%
21%–25%	1399	6057	18.76%
≥26%	2015	7859	20.41%
Total	11,298	49,077	

database funded by the National Highway Traffic Safety Administration that includes EMS information from 43 states and 2 territories (Guam and Virgin Islands) [12]. NEMSIS data originates from the patient care reports completed by emergency medical personnel from emergency activations and is submitted by local EMS agencies [13]. Due to the NEMSIS database providing de-identified information, this study is exempt from review from the institutional review board.

The NEMSIS database has information on a total of 74,231,756 EMS activations. This study only analyzed EMS cases that included cardiac arrest with the resulting ROSC outcome and information on the poverty level of the region where the EMS activation occurred. This inclusion criterion left a total of 150,003 cases to be the study population for this study. 73,886,409 cases were excluded from the study for having no recorded cardiac arrest and 195,344 cases were excluded from the study for having missing information regarding ROSC or poverty. Standard definitions were used from the NEMSIS database. ROSC was listed as the "Any Return of Spontaneous Circulation" in the NEMSIS database and was defined as "any brief (approximately >30 seconds) restoration of spontaneous circulation that provides evidence of more than an occasional gasp, occasional fleeting palpable pulse, or arterial waveform" [14]. Since the occurrence of ROSC was separated into two separate fields in the NEMSIS data labeled "Yes, At Arrival at the ED" and "Yes, Prior to Arrival at the ED", both fields were added together to create the total count of ROSC cases shown in Table 1 as "Yes ROSC". The SES of patients was measured by using the NEMSIS field "Geographic Region of Destination: Poverty – Percent". This field is defined by NEMSIS as the percentage of residents living in households with income below the federal poverty level in the zip code where the EMS activation occurred [14]. NEMSIS uses data from the 2014 American Community Survey from the US Census for the poverty percentage of each zip code in the US [14].

Since the poverty percentage returned from the NEMSIS database was in the form of intervals, chi-square testing was used to determine if there were statistically significant associations between different poverty intervals and cardiac arrest outcome variables. To measure the effect size of the associations between poverty intervals and cardiac arrest outcome variables, Pearson's r values were used to measure correlation due to the presence of interval data [15]. Linear regression coefficients were used to more precisely measure effect size. For all regressions, 95% confidence intervals were calculated. Data analysis was conducted using Microsoft Excel 2016 (Redmond, Washington) and R version 3.6.2 (Vienna, Austria). For all analyses, a p-value of <0.05 was used to signify statistical significance.

3. Results

A chi-squared test between poverty intervals and the amount of cardiac arrest cases that ended with no ROSC found a statistically significant association between the two variables ($p < 0.0001$). To further investigate this association, effect size was calculated through a Pearson's r value. To calculate the Pearson's r value, each poverty interval was simplified through the use of a numerical scale. The ≤5% poverty bracket was set as 1 and the 6%–10% poverty bracket was set as 2 with this pattern following until the highest poverty bracket (≥26%) was set as 6. This numerical scale was compared to the respective value of the percent of cardiac arrest cases that did not result in ROSC shown in Table 1 as "% No ROSC".

The resulting Pearson's r value was found to be 0.9424 ($R^2 = 0.8881$, $p < 0.005$). This indicates a strong positive correlation between increasing poverty level and an increasing chance that a cardiac arrest case does not result in ROSC [16]. Furthermore, a linear regression using the same input data from the Pearson's r calculation yielded a linear regression coefficient of 2.088 ($p < 0.05$, $R^2 = 0.8881$, 95% CI [1.059, 3.117]). This indicates that for every interval increase in poverty, there is an estimated 2% increase in the chance that a cardiac arrest case results in no recorded ROSC.

A chi-squared test between poverty intervals and amount of ROSC cases that were located in the ED revealed a statistically significant association between the two variables ($p < 0.0001$). To further investigate this association, effect size was calculated through a Pearson's r value. The same numerical scale from the previous Pearson's r calculation was compared to the percentage of ROSC cases that occurred in an ED which is shown in Table 2 as "% at ED" which yielded a Pearson's r value of 0.9005 ($R^2 = 0.8109$, $p < 0.05$). This indicates a strong positive correlation between increasing poverty level and an increased chance that ROSC occurs in the ED as opposed to prior to the ED [16]. Furthermore, a linear regression using the same input data from the Pearson's r calculation yielded a linear regression coefficient of 0.7701 ($p < 0.05$, $R^2 = 0.8109$, 95% CI [0.254, 1.286]). This indicates that for every interval increase in poverty, there is an estimated 0.77% increase in the chance that a cardiac arrest case that results in ROSC is located in the ED as opposed to in a prehospital location.

4. Discussion

The data not only indicates that individuals with a low SES in cardiac arrest have a lower chance of achieving ROSC but also that they have a lower chance of achieving ROSC prior to arrival at the ED. This suggests that even if individuals with a low SES are able to achieve ROSC, they face a higher chance of having a delayed ROSC at the ED instead of in a prehospital setting meaning that individuals with a low SES are more likely to incur neurological damage [17].

There are several possible explanations for why individuals with a lower SES have a lower chance of achieving ROSC and a higher chance of having delayed ROSC. First, there has been a link established between lower SES and longer EMS wait times in cases of cardiac arrest [18]. Longer EMS wait times can delay early CPR and defibrillation which have been shown to be critical to patient survival [19,20]. Furthermore, longer EMS wait times can delay the use of antiarrhythmic medications such as Amiodarone and Lidocaine which have been found to be less effective in later phases of cardiac arrest [21].

Secondly, bystander CPR is less likely in poor neighborhoods further delaying the start of CPR for low income individuals [22]. This is likely a result of a lack of CPR training as low income neighborhoods tend to have less CPR trained individuals compared to higher income neighborhoods [23]. Affordability is a major barrier for low income individuals to become CPR trained [24]. Thus, increases in funding for CPR training in low income neighborhoods is key to improving cardiac arrest survival outcomes. Unfortunately, current measures are not enough. The median annual CPR training rate for US counties is only 2.45% while many low income neighborhoods have an even lower rate of only 0.52% [23]. With survival rates of cardiac arrest cases doubling with the presence of bystander CPR, it is imperative that low income neighborhoods build up a presence of CPR trained individuals [25].

Thirdly, socioeconomic differences in cardiac arrest outcomes may be explained by low income individuals having a higher chance of multimorbidity resulting in a lower chance of ROSC occurring and a higher chance of delayed ROSC [26,27]. As a result, individuals with a lower SES are more likely to have worse cardiac issues [28].

It is worthwhile mentioning that low income individuals face an increased risk of cardiac arrest in the first place [29]. Individuals at the bottom income quartile in the US face double the risk of cardiac arrest compared to individuals in the top income quartile [30]. As a result, all the disparities faced in the treatment of cardiac arrest only serve to compound the original disparity in having higher rates of cardiac arrest. This shows that low income neighborhoods need to be the center point of increases in public health funding and improvements in cardiac arrest prevention and care.

Limitations of this paper include the fact that NEMSIS is a convivence sample and may not be a perfect nationally representative sample. Although there are over 21,000 licensed EMS agencies across the US, NEMSIS only has data from 9599 EMS agencies which may not be evenly distribution across different counties and states [12,31]. Moreover, information about individual patient income was not available so neighborhood poverty level was used as a substitute for measuring SES instead. Individual patient income would be a more precise indicator of SES than neighborhood poverty level as there can be significant differences in the SES of individuals within a zip code. Finally, the paper operates under the assumption that the zip code in which the EMS activation occurs is where the patient is located. However, this may not always be the case as patients can encounter cardiac arrest in zip codes in they do not live in with a drastically different socioeconomic makeup than their own.

5. Conclusion

There exist significant socioeconomic disparities in cardiac arrest outcomes in both the chance of ROSC occurring and the chance that ROSC is delayed. For every interval increase in poverty, the chance of an individual in cardiac arrest achieving ROSC decreases 2.09%. Furthermore, for every interval increase in poverty, the chance that ROSC is delayed until the ED as opposed to in a prehospital setting increases 0.77%. These socioeconomic disparities are only compounded by the fact that low income individuals already face an increased risk of cardiac arrest in the first place. Thus, this paper's findings suggest significant increases in funding are needed for public health measures in low income neighborhoods such as for CPR training to increase bystander CPR rates and increases in funding are needed for EMS services in low income neighborhoods to decrease EMS response times. Finally, more focus may be needed in low income neighborhoods in addressing the root causes of cardiovascular diseases to prevent cardiac arrests from happening in the first place.

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CRediT authorship contribution statement

Sriman Gaddam: Conceptualization, Formal analysis,Writing - original draft. **Sukhjit Singh:** Conceptualization, Writing - original draft.

Declaration of competing interest

None.

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